

Dr. Divyesh A. Mistry

Computational Materials Scientist | Multiscale Modeling | MD, DDD and HPC

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PROFESSIONAL SUMMARY

Computational materials scientist with more than five years of experience developing physics-based simulations of plasticity, defect evolution and fracture in advanced structural materials. Expertise spans molecular dynamics, discrete dislocation dynamics, polycrystalline microstructure modeling, high-performance computing and reproducible scientific software. Research applications include nickel-based superalloys, tungsten and tungsten-rhenium alloys under extreme thermomechanical environments.

TECHNICAL STRENGTHS

Methods: Molecular dynamics, discrete dislocation dynamics, multiscale mechanics, defect analysis, fracture and plasticity, microstructure-sensitive modeling

Software: LAMMPS, OpenDiS/ExaDiS workflows, OVITO, Neper, ParaView, Git, Linux/Unix, HPC schedulers

Programming: Python, NumPy, Pandas, scientific visualization, workflow automation, C/C++, Fortran

PROFESSIONAL EXPERIENCE

Visiting Researcher, Computational Materials Science

Merrimack College, North Andover, Massachusetts, USA | Apr 2026 - Jul 31, 2026

- Conducted multiscale MD and DDD research on crack-tip plasticity, defect evolution and ductile-to-brittle transition behavior in tungsten and tungsten-rhenium alloys.
- Developed automated LAMMPS and HPC workflows for temperature-dependent fracture simulations and large-scale post-processing.
- Connected atomistic fracture observations with mesoscale dislocation-network evolution for a forthcoming multiscale tungsten study.
- Collaborated on technical reports and research proposals and mentored students in computational mechanics and parallel molecular dynamics.

Junior Research Fellow, DRDO Project

Indian Institute of Technology Bombay, Mumbai, India | Jan 2025 - Dec 2025

- Investigated constitutive behavior and high-strain-rate response of shape-memory-alloy-based composites for aerospace and defense applications.
- Combined Split Hopkinson Pressure Bar experiments with numerical simulations for constitutive-model calibration and validation.
- Supported project execution, data analysis and mentoring in experimental characterization and computational mechanics.

Ph.D. Research Scholar

Indian Institute of Technology Bombay, Mumbai, India | Jan 2019 - Jun 2025

- Developed molecular dynamics and discrete dislocation dynamics models for defect mobility, precipitate strengthening and prior-particle-boundary effects in nickel-based superalloys.
- Built physics-informed workflows linking atomistic parameters with mesoscale dislocation mechanics using LAMMPS, Python and HPC systems.

Graduate Teaching Assistant

Indian Institute of Technology Bombay, Mumbai, India | Jan 2019 - Dec 2023

- Supported courses and computational laboratories in continuum mechanics, finite element methods and multiscale modeling.

Assistant Professor, Mechanical Engineering

CMR Institute of Technology, Bengaluru, India | Jun 2016 - Dec 2018

- Taught finite element methods, experimental stress analysis and solid mechanics, with computational modules using ANSYS, MATLAB and Python.

EDUCATION

Ph.D., Aerospace Engineering

Indian Institute of Technology Bombay | 2019 - 2025

- Thesis: Multiscale Modeling of Prior Particle Boundaries in Nickel-based Superalloys.

M.Tech., Mechanical Engineering - Mechanical Systems Design

Indian Institute of Technology Kharagpur | 2014 - 2016

SELECTED PUBLICATIONS AND SOFTWARE

A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-based Superalloys

Divyesh A. Mistry, Tawqeer Nasir Tak, R. Sankarasubramanian and P. J. Guruprasad

Journal of Engineering Materials and Technology, published online 2026. DOI: 10.1115/1.4072448

Size-dependent Power Laws for Edge Dislocations in Nickel Superalloys: A Molecular Dynamics Study

Divyesh A. Mistry and Amuthan A. Ramabathiran

Computational Materials Science 259 (2025), 114122. DOI: 10.1016/j.commatsci.2025.114122

Multiscale fracture and collective dislocation evolution in tungsten-based materials

Divyesh A. Mistry and collaborators

Journal manuscript in preparation.

Polycrystalline Tungsten Discrete Dislocation Dynamics Workflow

Dr. Divyesh A. Mistry

Open-source modular Python workflow with a reproducible 15-grain example, tests and citation metadata.

SELECTED RECOGNITION

- Institution of Eminence travel support for TMS 2025, USA.
- IIT Bombay travel grant for COMPLAS 2023, Barcelona, Spain.
- MHRD Teaching Assistantship through research project, IIT Bombay.
- Graduate Aptitude Test in Engineering (GATE) 2014: All India Rank 69.
- Member, The Minerals, Metals & Materials Society; permanent associate member, Aeronautical Society of India.